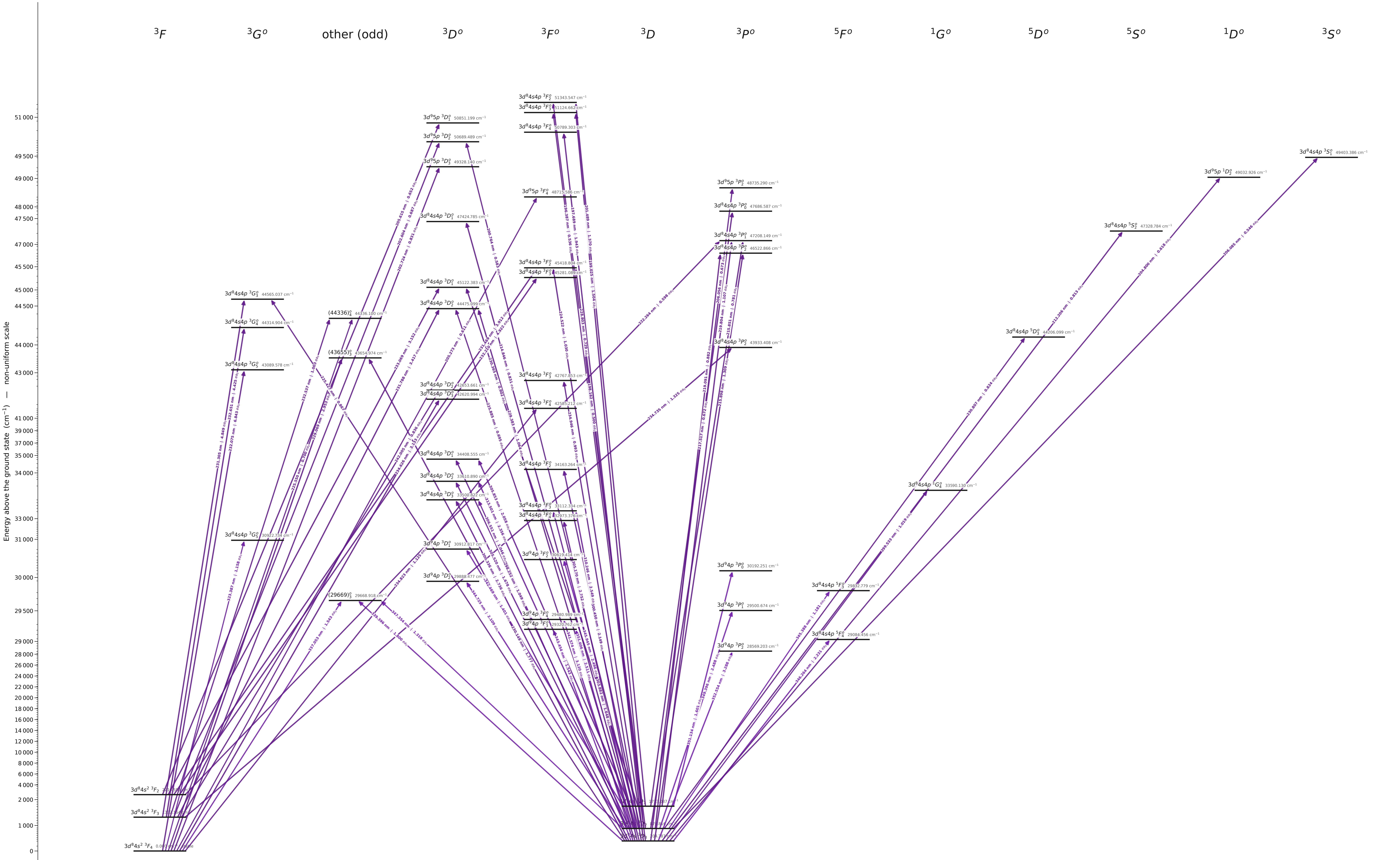
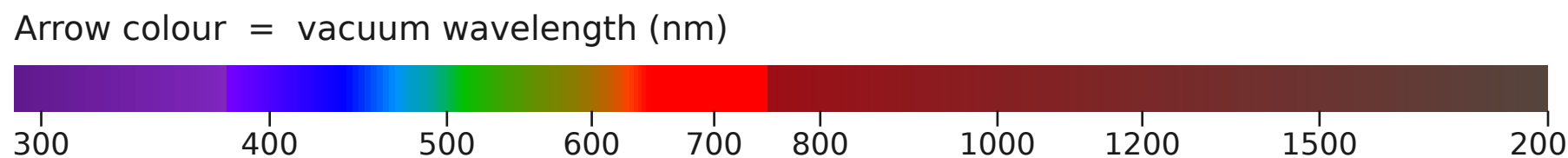


Nickel

Ni (Ni I). Strongest classified transitions below 2 μm (NIST): 55 levels, 70 lines.

Ni⁺ ionisation limit 61619.77 cm⁻¹ = 7.63988 eV (162.286 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U||er||U' \rangle$ in ea0
Matrix elements are measured or NIST values; * = high-accuracy theory, * = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U er U' \rangle$ (ea0)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3d ⁸ 4s 3D ₃ – 3d ⁸ 4s4p 3F ₂ ^o	197.6890	–	1516.4856	1.943	1.100e+08	–	–	–
3d ⁸ 4s 3D ₁ – 3d ⁸ 4s4p 3F ₂ ^o	201.4892	201.4242	1487.8838	1.370	9.300e+07	–	–	–
3d ⁸ 4s 3D ₁ – 3d ⁸ 4s4p 3P ₀ ^o	217.5166	217.4485	1378.2509	0.672	8.900e+07	–	–	–
3d ⁸ 4s ² 3F ₄ – (43655)§	229.0690	228.9984	1308.7432	2.953	2.100e+08	–	–	–
3d ⁸ 4s ² 3F ₂ – 3d ⁸ 4s4p 3G ₂ ^o	231.3055	231.2344	1296.0889	4.849	5.500e+08	–	–	–
3d ⁸ 4s ² 3F ₂ – 3d ⁸ 4s4p 3F ₂ ^o	231.4694	231.3983	1295.1710	3.912	5.000e+08	–	–	–
3d ⁸ 4s ² 3F ₂ – 3d ⁸ 4s4p 3D ₂ ^o	231.7877	231.7164	1293.3927	3.417	3.800e+08	–	–	–
3d ⁸ 4s ² 3F ₄ – 3d ⁸ 4s4p 3G ₂ ^o	232.0747	232.0034	1291.7930	6.843	6.900e+08	–	–	–
3d ⁸ 4s ² 3F ₂ – 3d ⁸ 4s4p 3F ₂ ^o	232.2096	232.1383	1291.0424	4.922	5.600e+08	–	–	–
3d ⁸ 4s ² 3F ₃ – 3d ⁸ 4s4p 3G ₂ ^o	232.6515	232.5801	1288.5901	4.425	3.500e+08	–	–	–
3d ⁸ 4s ² 3F ₂ – 3d ⁸ 4s4p 3D ₂ ^o	233.0685	232.9970	1286.2845	3.152	5.300e+08	–	–	–
3d ⁸ 4s ² 3F ₄ – 3d ⁸ 4s4p 3D ₂ ^o	234.6262	234.5543	1277.7453	3.133	2.200e+08	–	–	–
3d ⁸ 4s 3D ₁ – 3d ⁸ 4s4p 3D ₁ ^o	305.8528	305.7639	980.1855	2.058	1.000e+08	–	–	–
3d ⁸ 4s 3D ₂ – 3d ⁸ 4p 3P ₁ ^o	349.3955	349.2956	858.0317	2.488	9.800e+07	–	–	–
3d ⁸ 4s 3D ₁ – 3d ⁸ 4p 3P ₀ ^o	351.1339	351.0335	853.7639	1.601	1.200e+08	–	–	–
3d ⁸ 4s 3D ₃ – 3d ⁸ 4p 3P ₂ ^o	352.5544	352.4536	850.3438	3.288	1.000e+08	–	–	–

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).